

TCP vs UDP — Detailed Interview Notes

Introduction

Technical Definition

Transmission Control Protocol

TCP (Transmission Control Protocol) is a connection-oriented transport layer protocol that provides reliable, ordered, and error-checked data delivery between devices.

Technical Definition

User Datagram Protocol

UDP (User Datagram Protocol) is a connectionless transport layer protocol that sends data without establishing a connection and without guaranteeing delivery.

Why Transport Layer Protocols are Needed?

When data travels over network:

- Data may get lost
- Data may arrive out of order
- Data may get corrupted

Transport layer protocols solve these problems.

Main protocols:

- TCP
- UDP

Both work at:

- Transport Layer (OSI Layer 4)
-

TCP (Transmission Control Protocol)

TCP focuses on:

- Reliability
- Accuracy
- Ordered delivery

TCP ensures data reaches destination correctly.

Important Features of TCP

1. Connection-Oriented

Before sending data:

- Connection established
- Communication starts
- Connection terminated afterward

This process is called:

TCP Three-Way Handshake.

TCP Three-Way Handshake

Used to establish reliable connection.

Steps:

Step 1 — SYN

Client sends:

"SYN"

Meaning:

"I want to establish connection."

Step 2 — SYN-ACK

Server replies:

"SYN + ACK"

Meaning:

"Connection request accepted."

Step 3 — ACK

Client sends:

"ACK"

Connection established.

Visualization

Client → SYN

Server → SYN ACK

Client → ACK

Connection starts.

Why Handshake Needed?

Ensures:

- Both devices ready
 - Reliable communication possible
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2. Reliable Communication

TCP guarantees delivery.

If packet lost:

- TCP retransmits packet

This makes TCP reliable.

3. Ordered Delivery

Packets may travel through different paths.

TCP rearranges them correctly.

Example:

Message:

HELLO

Even if packets arrive:

- HLOEL

TCP rearranges:

HELLO

4. Error Checking

TCP checks:

- Missing packets
- Corrupted packets

Uses:

Checksum mechanism.

5. Flow Control

Prevents sender from overwhelming receiver.

Uses:

Sliding Window Protocol.

6. Congestion Control

Controls network traffic during congestion.

Avoids network overload.

Applications Using TCP

Because TCP is reliable, it is used where accuracy is important.

Examples:

- Web browsing
- Email
- File transfer
- Banking systems

Protocols using TCP:

- HTTP
 - HTTPS
 - FTP
 - SMTP
-

UDP (User Datagram Protocol)

UDP focuses on:

- Speed
- Low delay

UDP does NOT guarantee delivery.

Features of UDP

1. Connectionless

No connection establishment.

Packets sent directly.

No handshake.

2. Faster Communication

No reliability checking.

No acknowledgments.

Therefore:

Very fast.

3. No Ordered Delivery

Packets may arrive:

- Out of order
- Lost
- Duplicated

UDP does not fix these problems.

4. No Error Recovery

If packet lost:

UDP ignores it.

No retransmission.

Applications Using UDP

UDP used where:

- Speed more important than accuracy

Examples:

- Video streaming
 - Online gaming
 - Live broadcasting
 - Voice calls
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Why UDP Used in Video Calls?

Suppose one packet lost in video call.

Better to skip packet than delay whole video.

Therefore:

UDP preferred.

TCP vs UDP — Major Differences

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable	Unreliable
Speed	Slower	Faster
Error Checking	Yes	Minimal
Ordered Delivery	Yes	No
Acknowledgment	Yes	No
Flow Control	Yes	No
Congestion Control	Yes	No

Real Life Analogy

TCP Analogy

Courier service requiring signature.

Ensures package delivered correctly.

UDP Analogy

Radio broadcasting.

No confirmation whether listener received message.

Important Interview Questions

Difference between TCP and UDP?

TCP is reliable and connection-oriented.

UDP is fast and connectionless.

Why TCP slower?

Because it performs:

- Handshake
 - Error checking
 - Acknowledgment
 - Retransmission
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Why UDP faster?

No connection setup or reliability mechanisms.

Which applications use UDP?

Gaming, streaming, VoIP.

Which applications use TCP?

Web browsing, email, file transfer.

OSI vs TCP/IP — Detailed Interview Notes

Introduction to Network Models

Network communication is complex.

Models help standardize communication.

Two important models:

- OSI Model
 - TCP/IP Model
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OSI Model

Technical Definition

OSI Model

OSI (Open Systems Interconnection) model is a conceptual framework developed to standardize networking communication into seven layers.

Developed by:

- ISO organization

Purpose of OSI Model

Helps:

- Understand networking
- Standardize communication
- Troubleshoot networks

7 Layers of OSI Model

1. Physical Layer

Responsible for:

- Transmission of bits

Devices:

- Hub
- Repeater

2. Data Link Layer

Responsible for:

- Framing
- MAC addressing
- Error detection

Devices:

- Switch
- Bridge

3. Network Layer

Responsible for:

- Routing
- IP addressing

Device:

- Router

Protocols:

- IP
-

4. Transport Layer

Responsible for:

- Reliable delivery
- Segmentation

Protocols:

- TCP
 - UDP
-

5. Session Layer

Responsible for:

- Managing sessions
-

6. Presentation Layer

Responsible for:

- Encryption
 - Compression
 - Data translation
-

7. Application Layer

Provides user services.

Protocols:

- HTTP
 - FTP
 - SMTP
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TCP/IP Model

Technical Definition

TCP/IP

TCP/IP model is the practical networking model used on the Internet.

Developed by:

- DARPA
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Layers of TCP/IP

1. Network Access Layer

Combines:

- Physical
 - Data Link
-

2. Internet Layer

Equivalent to:

- Network Layer

Protocol:

- IP
-

3. Transport Layer

Protocols:

- TCP
 - UDP
-

4. Application Layer

Combines:

- Session
 - Presentation
 - Application
-

OSI vs TCP/IP Differences

Feature	OSI	TCP/IP
Layers	7	4
Type	Conceptual	Practical
Developed By	ISO	DARPA
Usage	Reference model Internet communication	
Session Layer	Separate	Combined
Presentation Layer	Separate	Combined

Why TCP/IP More Popular?

Because:

- Used on Internet
- Simpler
- Practical implementation exists

OSI mostly used for:

- Learning
 - Troubleshooting
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Important Interview Questions

Difference between OSI and TCP/IP?

OSI has 7 layers and is conceptual.
TCP/IP has 4 layers and is practical.

Which model used on Internet?

TCP/IP.

Which layer performs routing?

Network Layer.

Which layer uses TCP and UDP?

Transport Layer.

HTTP vs HTTPS — Detailed Interview Notes

HTTP

Technical Definition

Hypertext Transfer Protocol

HTTP (Hypertext Transfer Protocol) is an application layer protocol used for communication between web browsers and web servers.

Purpose of HTTP

HTTP transfers:

- Web pages
- Images
- Videos
- Data

between:

- Client (browser)
 - Server
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How HTTP Works?

1. User enters URL
 2. Browser sends HTTP request
 3. Server processes request
 4. Server sends HTTP response
 5. Browser displays webpage
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Problem with HTTP

HTTP sends data:

- In plain text

Meaning:

Anyone intercepting network can read data.

Dangerous for:

- Passwords

- Banking
 - Personal information
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HTTPS

Technical Definition

HTTPS

HTTPS (Hypertext Transfer Protocol Secure) is a secure version of HTTP that uses SSL/TLS encryption for secure communication.

SSL/TLS

Technical Definition

Transport Layer Security

TLS/SSL provides:

- Encryption
 - Authentication
 - Secure communication
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Why HTTPS is Secure?

HTTPS encrypts data before transmission.

Even if attacker captures packets:
Cannot read actual data.

HTTPS Security Features

1. Encryption

Data converted into unreadable form.

2. Authentication

Ensures website is genuine.

3. Data Integrity

Prevents modification during transmission.

HTTPS Working

1. Browser requests secure connection
 2. SSL/TLS handshake occurs
 3. Encryption keys exchanged
 4. Secure communication begins
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HTTP vs HTTPS Differences

Feature	HTTP	HTTPS
Security	No	Yes
Encryption	No	Yes
Port	80	443
Certificate	Not required	SSL certificate required
Speed	Slightly faster	Slightly slower
Usage	Normal sites	Secure websites

Real Life Analogy

HTTP

Sending postcard.

Anyone can read message.

HTTPS

Sending locked envelope.

Only receiver can read message.

Important Interview Questions

Difference between HTTP and HTTPS?

HTTPS provides encrypted secure communication using SSL/TLS.

Why HTTPS important?

Protects sensitive data.

What is SSL/TLS?

Security protocols used for encryption.

Which port HTTPS uses?

443.

Which layer HTTP works on?

Application Layer.